

 An official website of the United States government [Here's how you know](#)

[Welcome Juliette Bruce](#) | [Sign Out \(Home\)](#) | [My Profile](#) | [Contact](#)

Proposals	Reviews & Meetings	Awards & Reporting	Fellowships & Opportunities	Manage Financials	Administration
-----------	--------------------	--------------------	-----------------------------	-------------------	----------------

Proposal Review 3 : 2502380

[Back to Proposal](#)

Agency Name:	National Science Foundation
Agency Tracking Number:	2502380
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Tropicalizations of Shimura Varieties
Rating:	Good

Review

Summary

Proposal Synopsis

The PI proposes to study the tropical geomety of Shimura varieties, and to continue her successful record of conference organization and mentorship.

Intellectual Merit

Strengths

- The proposed project on Shimura varieties is a very natural line of inquiry, and the PI clearly has the technical chops to carry it out, given her impressive results on the weight zero cohomology of the moduli space of abelian varieties.
- The PI is clearly a very strong mathematician with a broad background and impressive publication record.
- I find question 5.5 especially interesting and unique due to the potential connections with number theory.
- Intellectual merit rating: G

Weaknesses

- The proposal is one-dimensional: besides for the discussion of prior work, the proposal is entirely focused on the weight-zero cohomology of Shimura varieties. This is certainly an interesting problem deserving of funding, but I would have liked to see some discussion of other directions the PI would like to go in as well. To give one example, the connection to number theory in question 5.5 could have been fleshed out more.
- The weight-zero direction continues a program initiated by the PI's postdoc advisor and their collaborators. There is nothing at all wrong with this, but the proposal under review does not include the development of an individual research program, unlike several other proposals under review.

Broader Impacts

Strengths

- The PI's prior organizational work supporting women and gender minorities in mathematics is very inspirational, and it merits financial support.
- In particular, the high level of effort she puts into organizing inclusive conferences is extremely admirable and should be emulated by many more mathematicians. Conferences are one of main ways that mathematicians meet each other and discuss their ideas. It is extremely important that they are spaces where mathematicians of all identities feel included and safe to sha ideas and ask questions. Otherwise, conferences can be discouraging to early career mathematicians from underrepresented groups.
- The PI's idea to start a mentorship program for LGBTQ+ students applying to graduate school is a good one, and will help increase participation of underrepresented groups in mathemat
- The PI hopes to bring small groups of women and underrepresented minorities in mathematics together at SLMath to brainstorm ways to increase equity in mathematics. Such retreats se a good incubators for potential broader impacts.
- Broader impacts rating: V/G

Weaknesses

- The proposal is lacking in specific details about activities that the grant might support. I think the PI has a lot of excellent ideas for future broader impacts, but they are not fully fleshed out proposal. Here is one example: the idea for an REU program specifically for LGBTQ+ students is an excellent one that certainly should be pursued, but almost no details are provided abo steps the PI would take to make this happen (e.g. who else the PI is working with on making this happen, how the PI would find potential mentors, how a location would be chosen, obtaini funding, etc.).
- Let me emphasize that I do not at all doubt that the PI could make these activities happen, but some other proposals I am reviewing have provided a lot more detail about how proposed t impacts activities will be carried out.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

As for results from prior NSF support (postdoctoral fellowship), I rate the intellectual merit to be V and broader impacts to be E.

Summary Statement

The proposed direction on the tropical geometry of Shimura varieties is very interesting, and the PI has a great record of both research and service that I am confident she will continue. H since the research focus is narrow compared to some other proposals I am reviewing, and some of the broader impacts of the proposal are lacking in specificity, I can only rate this propos 'Good'. The PI is a promising, talented early-career researcher whom I admire, but I do not feel that this proposal is completely reflective of her excellence.

Of the 7 proposals for standard grants I have reviewed[Material redacted per NSF Policy], this one is ranked 7th.

Of all of the proposals I have reviewed[Material redacted per NSF Policy], this one is ranked 9th.

About Services	Contact	News & Discoveries	Funding & Awards	Publications & Abo
Account Management	Contact Service Desk	News	Recently Announced Funding Opportunities	NSF
Award Cash Management		Discoveries	Upcoming Funding Opportunity Due Dates	Publications
Service (ACMS)		Multimedia Gallery	A-Z Index of Funding Opportunities	About the National Science
Notifications & Requests			Find Funding	Foundation
Project Reports			Award Search	Careers
Proposal Status			Proposal & Award Policies & Procedures Guide (PAPPG)	Staff Directory
Public Access				

Feedback ▶

See all NSF social media ▶